

NEWSROOM PORTAL

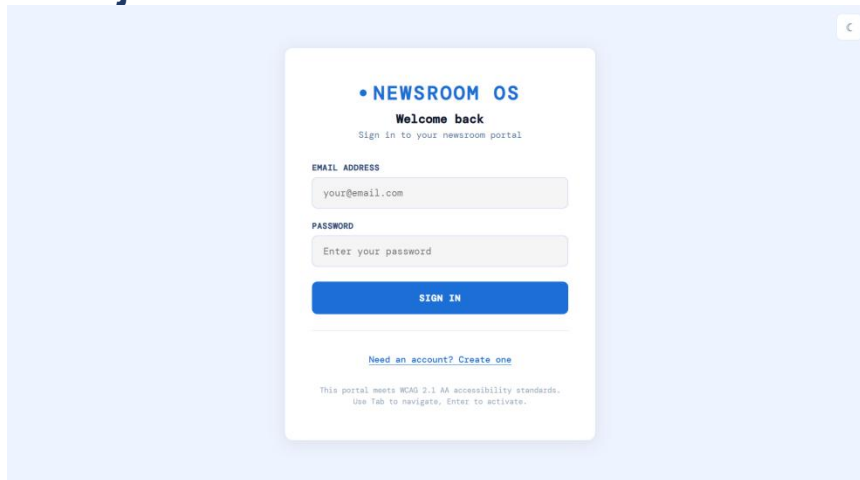
Assignment & Leave Management System

Complete Project Documentation | Version 2.0

Submitted to: Wipro Technologies | June 2026

Project Name	Newsroom Portal - Assignment & Leave Management
Technology Stack	React + TypeScript + Supabase + Groq AI + ffmpeg.wasm
Database	PostgreSQL (Supabase Cloud)
AI Models	Groq LLaMA 3.3 70B + Whisper Large v3
Repository	https://github.com/Architnaman/newsroom-portal
Cloud Database	https://vhremychmjzpunymvopt.supabase.co
Live URL	Netlify (gleaming-speculoos-44c554) — Production
Status	Production Ready

1. Project Overview



The Newsroom Portal is a full-stack, enterprise-grade, role-based web application designed to streamline story assignment, reporter availability management, leave request workflows, real-time team communication, AI-assisted operations, and management analytics for a newsroom organization. The system supports three distinct user roles — Admin, Editor, and Reporter — each with a tailored interface and database-level permission set enforced via Supabase Row Level Security.

The application integrates an AI-powered chatbot using the Groq LLaMA 3.3 70B model and an AI meeting report generator (Ambient Scribe) using Groq Whisper Large v3 for voice transcription, enabling natural language interactions and automated meeting documentation.

1.1 Key Objectives

- Automate story assignment using an AI scoring engine based on beat match, availability, and workload
- Provide real-time visibility into reporter availability and capacity across the organization
- Streamline leave request and approval workflows between reporters and editors
- Enable AI-powered meeting transcription and report generation via Ambient Scribe
- Support real-time team chat with DMs, group channels, reactions, and file attachments
- Deliver a complete email notification system covering all 9 key workflow events
- Provide workflow and app usage analytics dashboards for editors and administrators
- Allow reporters to record and submit video/audio reports directly from the portal
- Enable holiday-aware assignment with override workflows for public holidays
- Ensure cross-platform accessibility via GitHub Codespaces with no installation required

2. Technology Stack

Technology	Version	Purpose
React + TypeScript	18.x / 5.x	Frontend framework with type safety and hooks
Vite	5.x	Build tool and development server
Supabase JS Client	2.x	Database, auth, real-time subscriptions, storage
PostgreSQL	15.x	Primary relational database via Supabase
Groq LLaMA 3.3 70B	Latest	AI chatbot, report generation, transcript merging
Groq Whisper Large v3	Latest	Voice-to-text transcription for meeting recordings
React Router	6.x	Client-side routing and navigation
Recharts	Latest	Data visualization for analytics dashboards
ffmpeg.wasm	0.12.15	Browser-side video trim and multi-clip merge for reporter video submissions
Supabase Edge Functions	Deno	Reporter scoring algorithm and email notification dispatch
Supabase Storage	S3-compatible	Story files, chat attachments, video/audio reports, PDF guides
Gmail SMTP + denomailer	Latest	Email notification delivery for all 9 workflow event types
Netlify	Free tier	Frontend hosting, CI/CD via GitHub main branch
GitHub Codespaces	Cloud	No-install deployment for restricted environments

3. System Architecture

The application follows a modern three-tier architecture with a React frontend, Supabase as the backend-as-a-service platform, and Groq AI for intelligent automation. Real-time updates are delivered via Supabase PostgreSQL Change Data Capture (CDC) subscriptions.

Layer	Technology	Description
Frontend	React 18 + TypeScript	UI, state management, theming, routing, context API
Authentication	Supabase Auth	JWT-based auth with role-based access (editor/reporter/admin)
Database	Supabase PostgreSQL 15	20 tables with RLS, triggers, functions
Real-time	Supabase Realtime	Live chat, typing indicators, dashboard refresh via WebSocket
AI Chatbot	Groq LLaMA 3.3 70B	Natural language commands with full DB context and action execution
AI Transcription	Groq Whisper Large v3	Meeting voice recording to text with multi-accent support
Edge Functions	Deno (Supabase)	Reporter scoring: beat match + availability + headroom; email dispatch
File Storage	Supabase Storage (S3)	Story files, chat attachments, video/audio reports, PDF guides
Notifications	Gmail SMTP / denomailer	9-type email system firing from all assignment surfaces
Analytics	Recharts + page_events table	Workflow analytics and app usage tracking
Video Processing	ffmpeg.wasm	Client-side video trim and multi-clip merge (no server needed)

4. Database Schema

The application uses 20 PostgreSQL tables across three logical groups. All tables have Row Level Security enabled with appropriate per-role policies enforced at the database level.

4.1 Core Tables (11)

Table	Purpose	Key Columns
reporters	Reporter profiles	id, name, email, beats[], complexity_level, max_stories_per_week, last_seen_at
stories	News stories	headline, category, urgency, complexity, priority, deadline, status
assignments	Story-reporter links	story_id, reporter_id, is_override, override_status, override_response
availability	Weekly availability	reporter_id, week_start_date, available_days[]
leave_requests	Leave tracking	reporter_id, leave_date, leave_type, status, filed_by_editor, editor_note
profiles	Auth role mapping	id (auth.users), role (editor/reporter/admin), reporter_id
notification_log	Notification audit trail	reporter_id, story_id, type, recipient_email, email_status, error_message
holidays	Public holidays	date, name, is_recurring (US federal holidays 2026)
admins	Admin records	id (auth.users), name, email
app_settings	Portal settings	key (date_format, week_start_day), value
leave_filing_requests	Editor-filed leaves	reporter_id, requested_date, reason, status, editor_note

4.2 AI and Analytics Tables (2)

Table	Purpose	Key Columns
ai_reports	Ambient Scribe reports	story_ids[], transcript, story_notes, assignment_notes, rostering_notes, confidence_score, status
page_events	App usage analytics	user_id, reporter_id, role, event_type, page_path, session_id, created_at

4.3 Team Chat Tables (6)

Table	Purpose	Key Columns
chat_channels	DM and group channels	id, name, type (dm/group), created_by, is_archived
chat_channel_members	Channel membership	channel_id, reporter_id, last_read_at, is_admin

chat_messages	All chat messages	channel_id, sender_id, content, reply_to_id, is_pinned, is_deleted
chat_reactions	Message emoji reactions	message_id, reporter_id, emoji
chat_typing	Typing indicators	channel_id, reporter_id, updated_at
chat_attachments	File attachments	message_id, file_url, file_name, file_type, file_size

4.4 Database Functions and Triggers

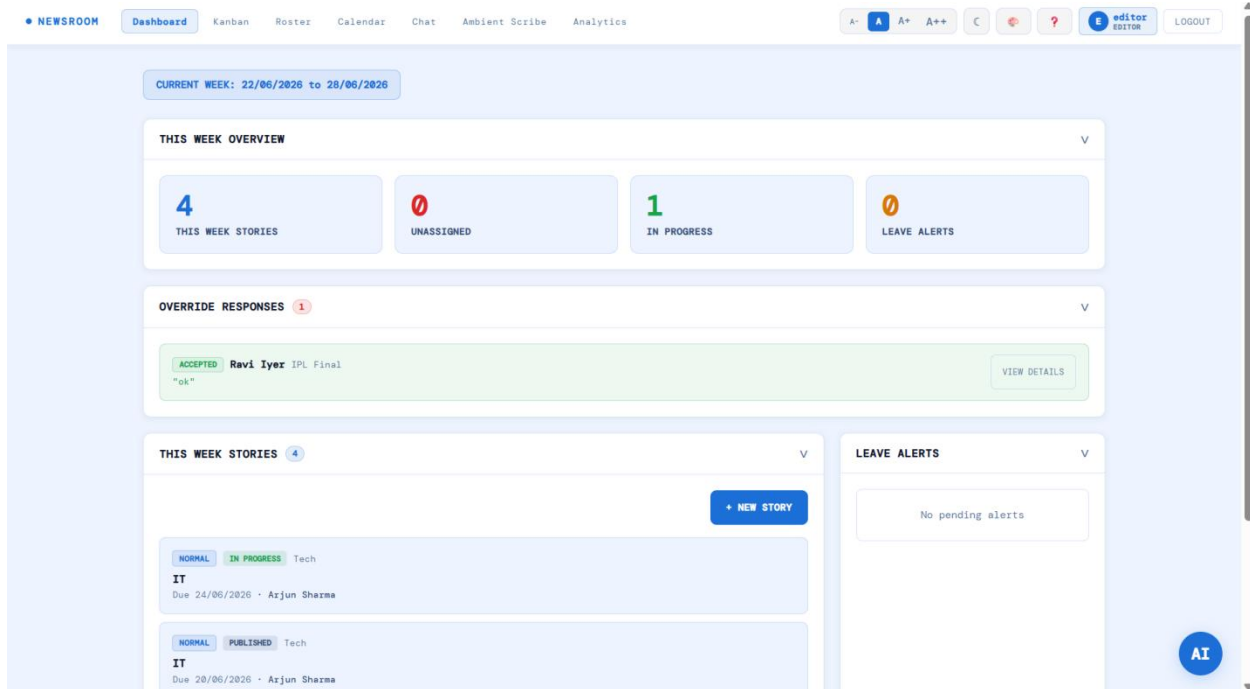
- auto_update_complexity() — trigger that automatically recalculates reporter complexity_level when a story they are assigned to is filed or published
- reset_weekly_reporter_load() — deactivates assignments for filed/published stories to reset weekly capacity
- is_channel_member() — SECURITY DEFINER helper function that prevents RLS infinite recursion on chat_channel_members queries

5. User Roles and Access Control

Role	Login Email	Password	Key Capabilities
Admin	admin@newsroom.com	admin123	Date format, week start day, system settings, all analytics
Editor	editor@newsroom.com	editor123	Create/assign/publish stories, approve leaves, Ambient Scribe, Analytics
Priya Mehta	priya@newsroom.com	editor123	Politics, Economy
Arjun Sharma	arjun@newsroom.com	editor123	Tech, Science
Fatima Nair	fatima@newsroom.com	editor123	Crime, Local
Ravi Iyer	ravi@newsroom.com	editor123	Sports, Entertainment
Sunita Rao	sunita@newsroom.com	editor123	Business, Economy

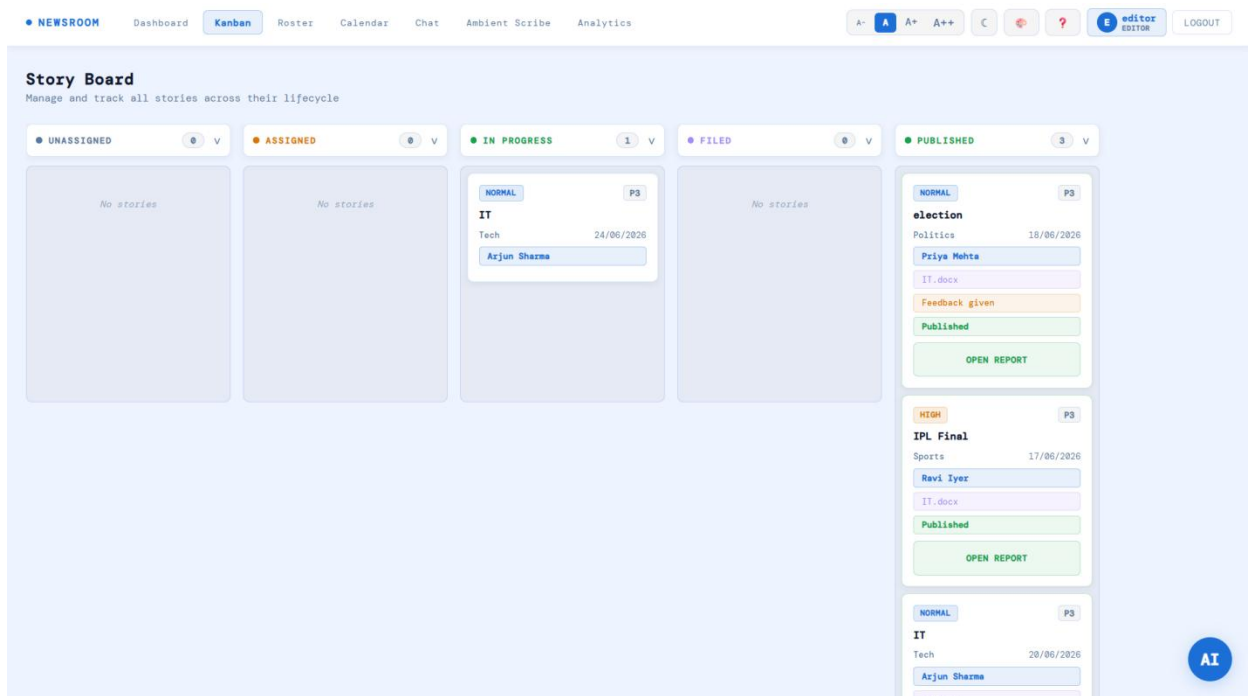
6. Features — Editor Portal

6.1 Editor Dashboard



- Summary cards: This Week Stories, Unassigned count, In Progress count, Leave Alerts
- Leave alerts panel showing pending and emergency leave requests with quick approve/reject
- New story creation with headline, category, urgency, complexity, priority, deadline
- Collapsible sections for organized workflow management
- Real-time updates via Supabase subscriptions — no page refresh needed

6.2 Kanban Board

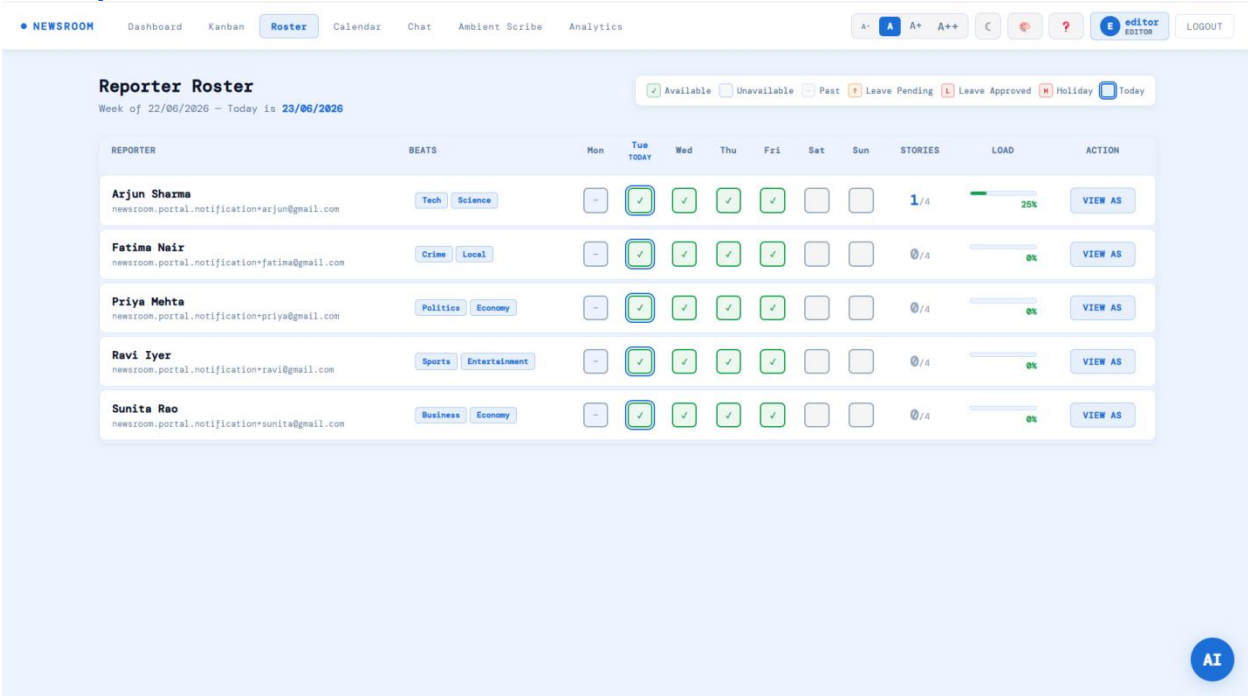


- Visual story pipeline: Unassigned → Assigned → In Progress → Filed → Published
- Collapsible columns with story count badges per status
- View and review filed stories with approve/reassign/reject actions
- Editor feedback on story publication sent directly to reporter via email
- Reassign stories back to unassigned with mandatory reason

6.3 AI-Powered Reporter Scoring and Assignment

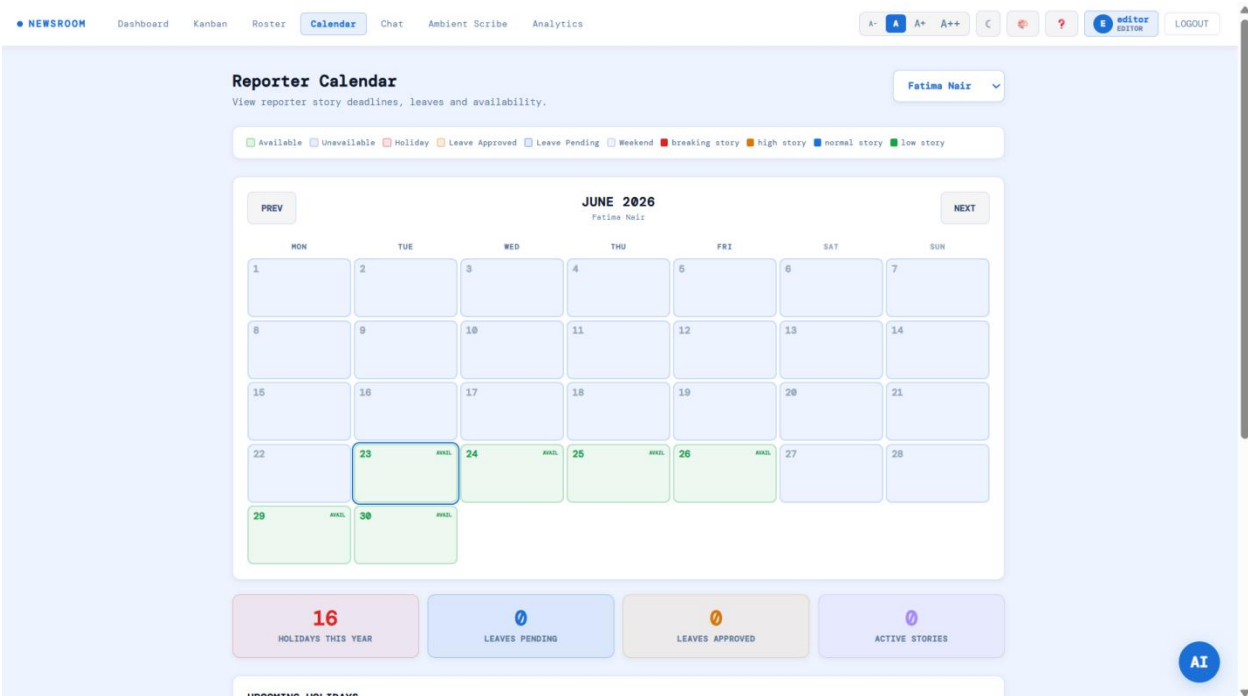
- AI scoring engine ranks reporters by beat match (40%), availability (35%), and headroom (25%)
- Supabase Edge Function (score-reporters) evaluates every active reporter against the story
- Top match highlighted with BEST badge and score out of 100
- Override assign available for unavailable reporters with mandatory reason field
- Holiday override — deadline on public holiday forces override workflow for ALL reporters

6.4 Reporter Roster



- Weekly availability grid showing all reporters across Mon-Sun
- Color-coded cells: Available (green), Leave Pending (yellow), Leave Approved (red), Holiday (red H)
- Story capacity bar showing current load vs maximum stories per week
- VIEW AS button to impersonate any reporter and view their full dashboard

6.5 Calendar View



- Monthly calendar with story deadlines, reporter leaves, and public holidays
- Reporter filter dropdown to view individual reporter calendars
- Week start day aligned to admin setting (Monday or Sunday)
- Holiday indicators displayed in red across all calendar cells

7. Ambient Scribe — AI Report Generator

NEWSROOM

Dashboard

Kanban

Roster

Calendar

Chat

Ambient Scribe

Analytics

A- A+ A++ C ? editor EDITOR LOGOUT

Ambient Scribe

Select stories → choose attendees → record or type transcript → AI validates reporters → generate report

Generate Report Saved Reports (1)

Step 1 – Select Stories

Step 2 – Select Meeting Attendees ***required**

Select reporters who will be in this meeting. This helps AI extract names more accurately.

FN Fatima Nair

Column: Local | 0/4 stories

SR Sumita Rao

Business, Economy | 0/4 stories

PM Priya Mehta

Politics, Economy | 1/4 stories

AS Arjun Sharma

Tech, Science | 2/4 stories

RI Ravi Iyer

Sports, Entertainment | 1/4 stories

Step 3 – Choose Transcript Method

How would you like to provide the meeting transcript?

Voice + Manual

Voice Only

Manual Only

AI

Ambient Scribe is the AI-powered meeting report generator built into the Editor portal. It converts editorial meeting discussions into structured, validated assignment reports using a combination of voice recording, manual transcription, and multi-layer AI extraction.

7.1 Complete Workflow

Step	Action	Description
1	Select Stories	Choose one or more stories to discuss. Only unassigned or assigned stories shown.
2	Select Attendees	REQUIRED: Select every reporter attending. Mandatory — enables accurate AI name matching.
3	Choose Transcript Mode	Voice + Manual, Voice Only, or Manual Only
4	Voice Recording	Click START RECORDING. Auto-gain control, echo cancellation and noise suppression enabled.
5	AI Transcription	Groq Whisper Large v3 transcribes audio. Supports all accents. Returns plain text in seconds.
6	Merge Transcripts	In Voice + Manual mode, LLaMA 3.3 70B merges both. Manual is primary; voice fills gaps.
7	Reporter Extraction	3-layer AI system extracts which reporter was mentioned for each story from the transcript.
8	Validation	Each extracted reporter validated: leave status, availability, capacity, beat match all checked.
9	Confirm Assignments	Editor confirms, changes, or overrides each story assignment with mandatory reason if overriding.
10	Generate Report	LLaMA 3.3 70B generates Story Notes, Assignment Notes and Rostering

		Notes sections.
11	Review and Edit	Editor reviews all three sections. Live confidence score updates as edits are made.
12	Approve and Assign	Stories assigned, report visible to all reporters, override reporters notified to accept/reject.

7.2 3-Layer AI Reporter Extraction

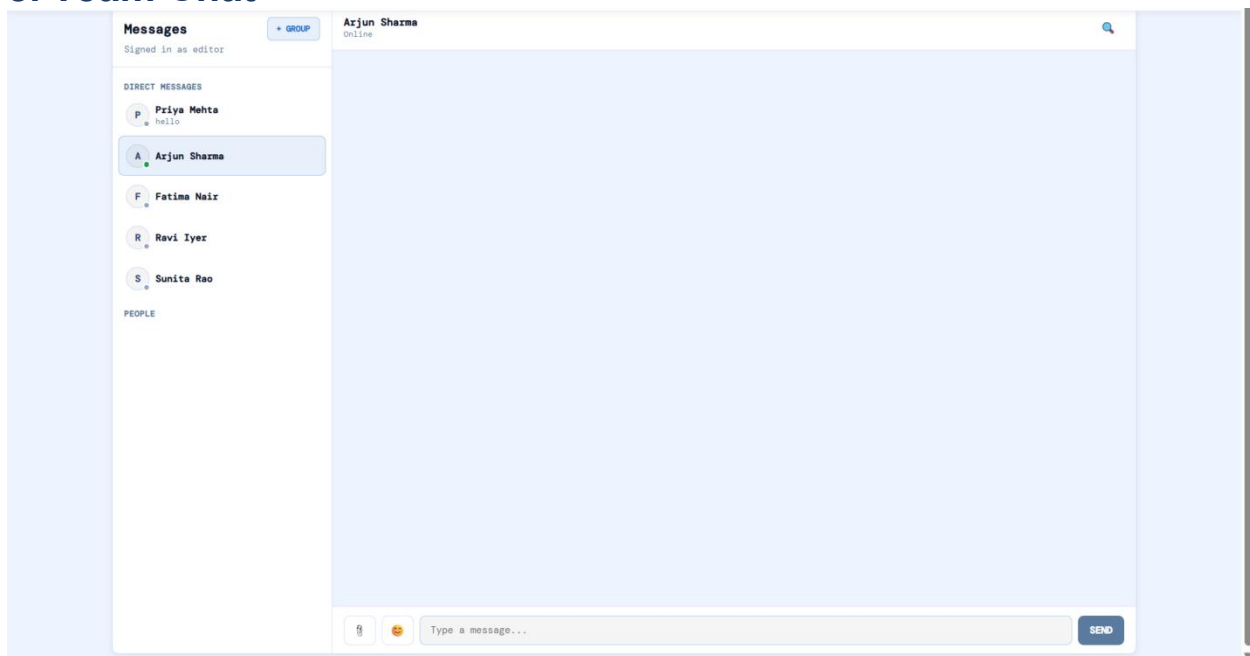
Layer	Method	Description
1	LLaMA 3.3 70B AI	AI reads full transcript with known reporter names and meeting attendees as context. Detects assignment language (assign, cover, handle, give it to).
2	Keyword Proximity Scan	Scans transcript for reporter first names appearing within 150 characters of story keywords. Scores each combination and picks highest.
3	Graceful Fallback	If both layers fail, returns empty and shows top 3 suggestions based on scoring engine. Editor picks manually.

7.3 Confidence Score

Check Point	Points	What It Verifies
Not on leave	25 pts	Reporter has no acknowledged leave on assignment date
Available	20 pts	Reporter has set availability this week
Capacity OK	20 pts	Reporter is not at maximum story limit
Beat match	20 pts	Reporter beats match the story category
Mentioned in report	15 pts	Reporter name appears in generated assignment notes

75% and above — Green (High confidence). 50-74% — Yellow (review carefully). Below 50% — Red (consider regenerating).

8. Team Chat



A full-featured real-time chat system built on Supabase Realtime, supporting direct messages and group channels across all user roles.

8.1 Features

- Direct Messages (DMs) — one-to-one conversations between any two users
- Group Channels — named channels with multiple members, admin roles, and @mention support
- Real-time typing indicators — shows who is currently typing in a channel
- Online / last-seen status — green dot for users active within last 2 minutes
- Message reactions — 8 emoji reactions per message with toggle support
- Reply threading — reply to any specific message with quoted preview
- Message pinning — pin important messages, shown in a banner at the top of the chat
- Message editing and deletion — edit your own messages, soft-delete with 'Message deleted' placeholder
- File and image attachments — upload any file type; images render inline
- @mention autocomplete — type @ in group channels to get a member dropdown
- Message search — search within any conversation by keyword
- Unread message badges — blue dot indicators on conversations with unread messages
- Seen receipts — last message shows 'Seen' when the other party has read it
- SECURITY DEFINER helper function (is_channel_member) to prevent RLS infinite recursion

9. Email Notification System

A complete email notification system built on Supabase Edge Functions and Gmail SMTP via the denomailer library. All notifications are dispatched from a dedicated Gmail account (newsroom.portal.notification@gmail.com) using a Google App Password. Every send is logged to the notification_log table with delivery status.

9.1 Notification Types

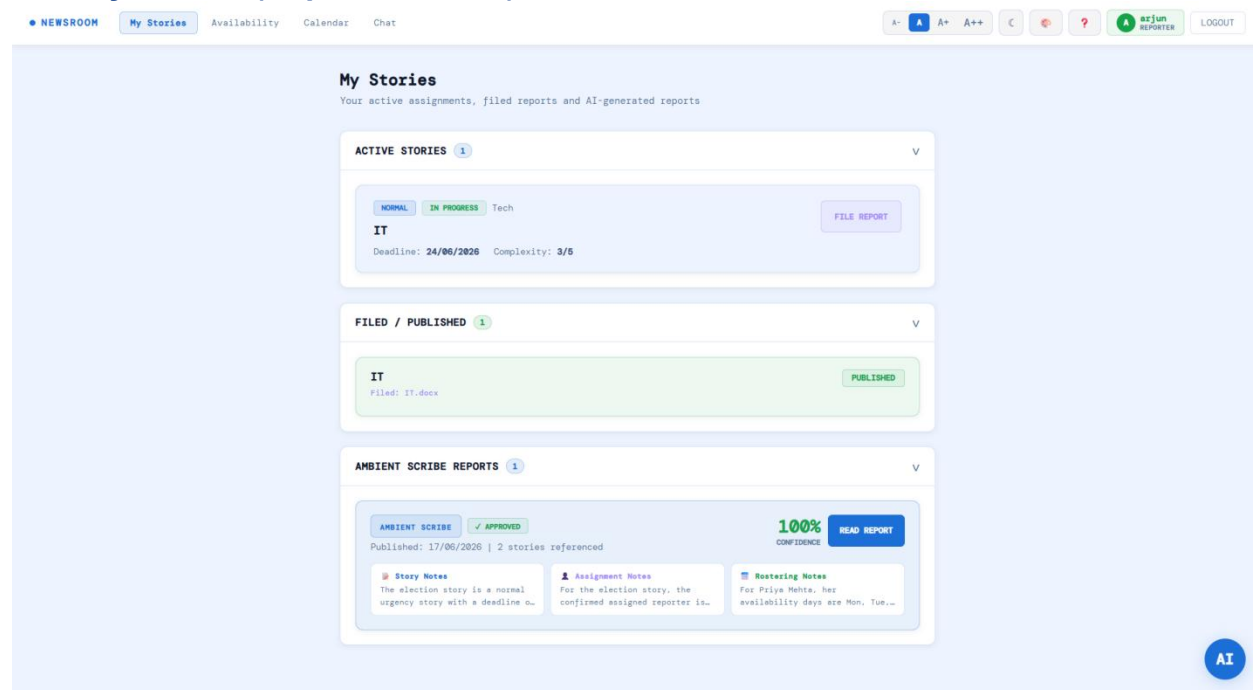
#	Event	Recipient	Trigger Surface
1	Story assigned	Reporter	Assign Modal, Chatbot, Ambient Scribe, Kanban
2	Override assignment (action required)	Reporter	All assignment surfaces
3	Story published	Reporter	Kanban Board, Chatbot
4	Story reassigned	Reporter	Kanban Board, Editor Dashboard
5	Override response (accepted/rejected)	Editor	Reporter Queue, Chatbot
6	Story filed by reporter	Editor	Reporter Queue file modal, Chatbot
7	Leave requested	Editor	Reporter Queue, Chatbot
8	Leave acknowledged	Reporter	Editor Dashboard, Chatbot
9	Leave rejected	Reporter	Editor Dashboard, Chatbot

9.2 Coverage

Notifications fire from all assignment surfaces — the manual Assign Modal, Kanban Board, AI Chatbot, and Ambient Scribe — ensuring complete coverage regardless of which workflow path is used. The notification_log table records recipient_email, email_status (sent/failed), and error_message for every send attempt.

10. Features — Reporter Portal

10.1 My Stories (Reporter Queue)



- View all active story assignments with headline, category, urgency, deadline, and complexity
- START WORKING button to mark story as In Progress
- FILE REPORT button opens the enhanced file submission modal for In Progress stories
- Override assignment alerts requiring ACCEPT or REJECT with a written reason
- AMBIENT SCRIBE REPORTS section showing all approved AI meeting reports from the editor
- READ REPORT button opens full three-section report: Story Notes, Assignment Notes, Rostrering Notes

10.2 Enhanced File Submission Modal

The file submission modal supports multiple methods of sourcing files to attach to a report:

- BROWSE — standard file picker supporting all file types, multiple file selection
- PHOTO — live camera capture with drag-to-crop and zoom controls, works on both desktop and mobile
- GALLERY — mobile gallery picker for selecting existing photos
- VIDEO — full video recorder with back camera default on mobile, front/back switch, recording timer, and live preview
- AUDIO — standalone audio-only recorder with animated waveform display
- Drag-and-drop — drop any files directly into the modal drop zone

Video Recording Features

- Records using device back camera by default (field reporting orientation), one tap to switch to front camera

- High-resolution recording: ideal 1920x1080 at 30fps with VP9/Opus codec
- Multi-clip recording — reporter can record multiple separate clips in one session
- Per-clip trim editor — drag range sliders to set exact trim start and trim end for each clip
- ffmpeg.wasm merge — all trimmed clips are concatenated into a single output video file using WebAssembly FFmpeg running entirely in the browser with no server required
- Progress bar during merge processing with estimated completion percentage
- Local video preview — merged video plays back inline before submission so reporter can verify quality
- Processed video is automatically added to the selected files list and uploads to Supabase Storage on Submit

Audio Recording Features

- Standalone microphone-only recording mode separate from video
- Animated waveform bars display during active recording
- Multiple audio clips supported in one session
- Per-clip inline audio player for playback before submission
- Each clip can be individually added to the file list or discarded

10.3 Availability Management

The screenshot shows a web application interface for 'Availability Management'. At the top, there's a navigation bar with links: NEWSROOM, My Stories, Availability (active), Calendar, and Chat. On the right, there are user controls: a search bar, a user profile icon labeled 'acjun REPORTER', and a LOGOUT button. The main content area is titled 'Availability' with a subtitle 'Manage your weekly availability and leave requests'. It features three main sections: 1. 'THIS WEEK'S AVAILABILITY' showing a grid for the week of 22/06/2026. The grid has columns for Mon (PMT), Tue, Wed, Thu, Fri, Sat, and Sun. Tue, Wed, and Thu are highlighted in green, indicating they are selected. A 'SAVE AVAILABILITY' button is at the bottom of this section. 2. 'LEAVE REQUESTS' showing a '+ FILE LEAVE' button and a message 'No leave requests filed yet'. 3. 'REQUEST EDITOR TO FILE LEAVE' showing a text input field and a '+ REQUEST' button. A blue circular 'AI' button is visible in the bottom right corner.

- Weekly day selector showing Mon-Sun availability grid
- Holiday blocking — public holidays shown as HOL in red and cannot be selected
- Leave-approved days shown in red, leave pending in yellow
- Default availability set to Mon-Fri excluding any holidays that fall that week
- Save button instantly updates database with real-time broadcast to editor roster

10.4 Leave Requests

My Calendar
View your story deadlines, leaves and availability. Tap any future date to apply for leave.

Available Unavailable Holiday Leave Approved Leave Pending Weekend breaking story high story normal story low story

JUNE 2026
Your calendar

PREV NEXT

MON	TUE	WED	THU	FRI	SAT	SUN
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18 NORMAL IT 2025-06-18 - 20/06/2025	19 NORMAL IT 2025-06-18 - 20/06/2025	20 NORMAL IT 2025-06-18 - 20/06/2025	21
22	23 NORMAL IT 2025-06-23 - 24/06/2025	24 NORMAL IT 2025-06-23 - 24/06/2025	25 AVAIL	26 AVAIL	27	28
29 AVAIL	30 AVAIL					

16 HOLIDAYS THIS YEAR 0 LEAVES PENDING 0 LEAVES APPROVED 2 ACTIVE STORIES

AI

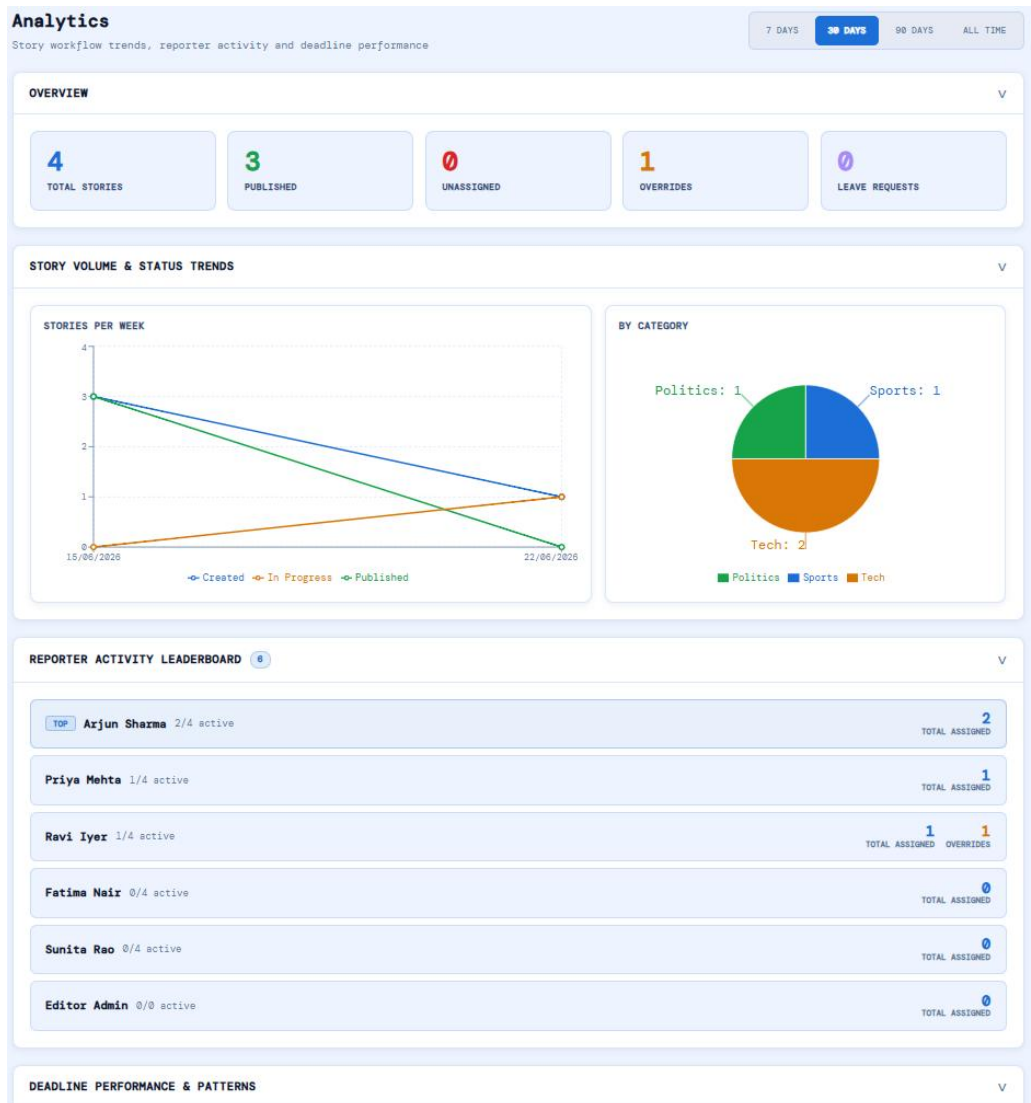
- File leave for any future date with type: Planned, Sick, or Emergency
- Emergency and sick leave immediately alerts editors with email notification
- View all leave history with status: Pending, Acknowledged, Rejected
- FILED BY EDITOR badge shown for editor-filed leaves
- Reporter can request editor to file leave on their behalf with a mandatory reason

11. Analytics Dashboards

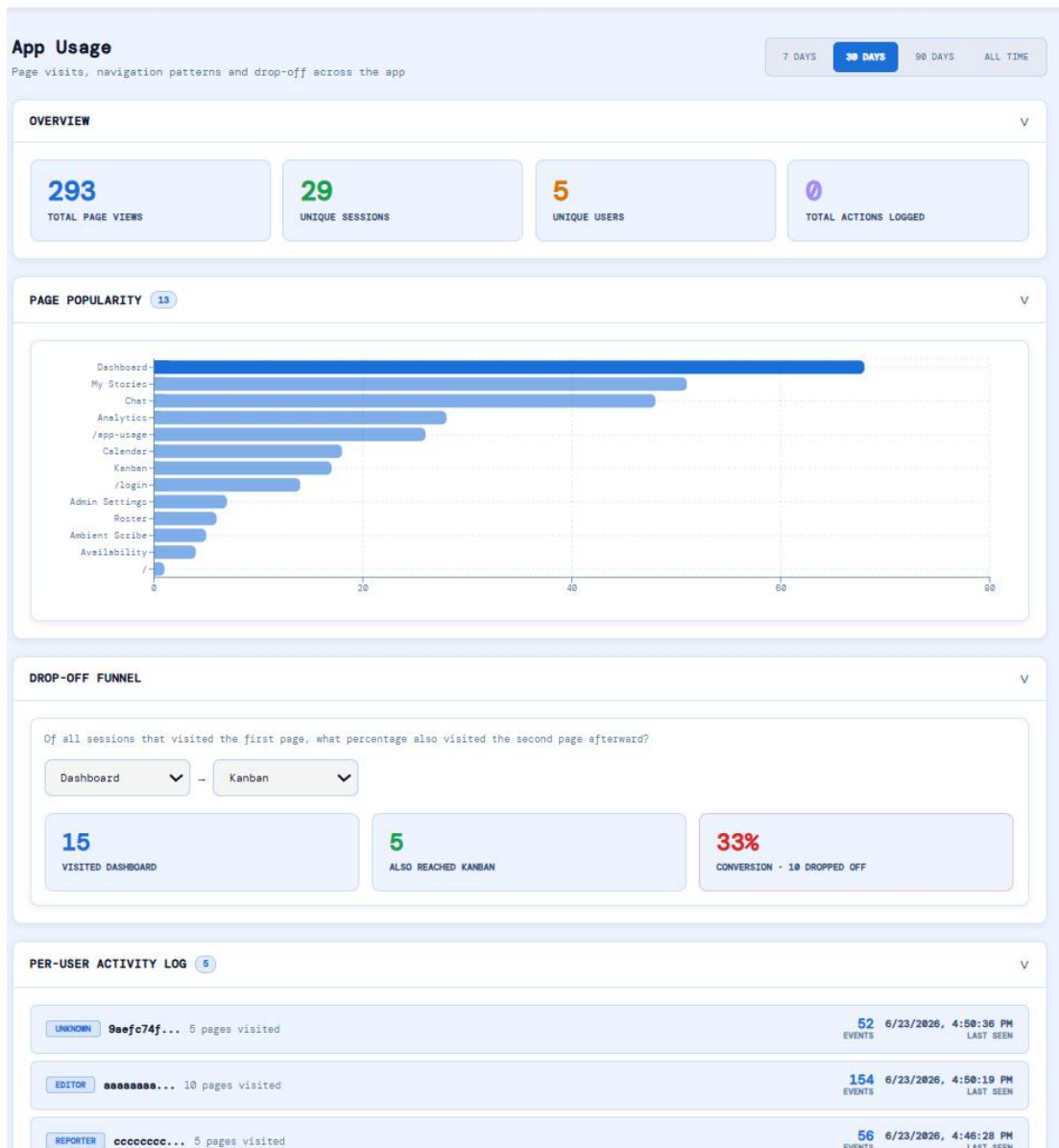
11.1 Workflow Analytics (Editor + Admin)

A dedicated Analytics page visible to editors and admins, built using Recharts, covering editorial output metrics with configurable date range filters (7 days / 30 days / 90 days / all time).

- Story volume and status trends over time — line chart grouped in weekly buckets
- Story category breakdown — pie chart showing distribution across all categories
- Reporter activity leaderboard — total assignments, override frequency, active vs capacity
- Deadline performance — on-time vs late filing rate displayed as a bar chart
- Override assignment patterns — percentage of assignments that required override (pie chart)
- Leave request patterns — total, pending, approved, rejected by type (bar chart)



11.2 App Usage Analytics (Admin only)



A Google Analytics-style usage tracking system built from scratch, visible only to the Admin role. Every route change logs a `page_view` event to the `page_events` Supabase table via a custom `usePageTracking` React hook.

- Page popularity ranking — total visits per page across all users, displayed as a horizontal bar chart. Each bar shows how many total page-load events occurred for that route across the selected period
- Drop-off funnel analysis — configurable from-page and to-page conversion rate. Shows what percentage of sessions that visited page A also subsequently visited page B in the same session. Example: 10 sessions visited Dashboard, only 2 reached Kanban — 80% drop-off rate, surfacing a discoverability problem
- Per-user activity log — every user's total events, unique pages visited, and last-seen timestamp. Identifies power users, inactive users, and onboarding gaps
- Configurable date range filter: 7 days / 30 days / 90 days / all time
- Data stored in a dedicated `page_events` table with RLS ensuring only admins can read it

12. AI Chatbot (Groq LLaMA 3.3 70B)

The AI Chatbot provides role-aware natural language interaction for both editors and reporters. It has full read access to database context including stories, reporters, leaves, and holidays — and can execute real database actions.

NEWSROOM AI
Editor Assistant

CLEAR

X

Hi Reporter! I can help you start working on stories, file leaves, check your stories and update availability. What would you like to do?

Type a message...

SEND

12.1 Editor Commands

Example Command	Action Executed
Create a breaking news story about the election with deadline tomorrow	Inserts new story into stories table
Assign the election story to Priya	Normal assign — creates active assignment, sends notification
Override assign to Arjun, he is on leave but it is urgent	Override assign — reporter must accept/reject, email sent
Approve Priya leave request	Updates leave status to acknowledged, sends notification

Publish the filed story with feedback great work	Updates story status to published with feedback, sends notification
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12.2 Reporter Commands

Example Command	Action Executed
Start working on my election story	Updates story status to in_progress
File a planned leave for next Monday	Creates leave_request with status pending, notifies editor
Accept the override assignment, I will cover it	Updates override_status to accepted, notifies editor
Reject the override, I am hospitalized	Updates override_status to rejected, story moved to unassigned

12.3 Holiday Awareness

- Chatbot fetches holidays on every interaction via getDBContext()
- Blocks normal assign_story when story deadline falls on a public holiday
- Forces override_assign_story with reason for holiday deadline assignments
- Warns reporter if they try to file leave on a public holiday date

13. Third-Party APIs and Services

All services are currently running on their respective Free Tier plans with no active paid subscriptions.

13.1 Groq API

Item	Details
Purpose	AI Chat, Report Generation, Voice Transcription
Models Used	llama-3.3-70b-versatile (chat + text generation), whisper-large-v3 (speech-to-text)
Used In	Chatbot.tsx (chat), AIReportGenerator.tsx (text + transcription)
Plan	Free Tier — no credit card required
LLaMA RPM / RPD / TPM	30 / 1,000 / 12,000
Whisper Requests/Day	2,000 / Audio seconds per day: 28,800
Risk	High AI traffic may exceed free-tier limits and temporarily throttle requests (HTTP 429)
Upgrade Path	Add credit card to unlock Developer tier (~10x limits) — billing only applies beyond free allowance

13.2 Supabase

Item	Details
Purpose	Database, Auth, Storage, Edge Functions, Realtime
Plan	Free Tier — no credit card required
Database Storage	500 MB limit
File Storage	1 GB limit
Monthly Active Users	50,000
Edge Function Invocations	500,000 / month
Bandwidth	5 GB / month egress
Critical Risk	Project auto-pauses after 7 days of zero API activity — app goes fully offline until manually resumed from dashboard
Upgrade Path	Pro plan \$25/month — removes inactivity pause, higher caps, daily backups

13.3 Gmail SMTP

Item	Details
Purpose	All 9 types of email notifications via Supabase Edge Function
Authentication	Google App Password (16-character scoped credential — not the real account password)
Sender Account	newsroom.portal.notification@gmail.com

Plan	Free standard Gmail account — no Google Workspace subscription
Daily Sending Limit	~500 emails/day (Google's standard free account limit)
Risk	Large email volumes may require migration to SendGrid or Resend

13.4 Netlify

Item	Details
Purpose	Frontend hosting, CI/CD — auto-deploys on push/merge to main branch
Plan	Free Tier — credits-based model (new account, 2026)
Monthly Credits	300 credits/month — hard limit, no overage
Cost per Deploy	15 credits per production deploy (~20 deploys/month maximum)
Cost per GB Bandwidth	20 credits/GB
Risk	Excessive deploys exhaust credits — site suspended until next billing cycle. Feature-branch git workflow used to conserve credits.
Upgrade Path	Personal \$9/month (1,000 credits), Pro \$20/month (3,000 credits)

14. Security Architecture

- Supabase Auth with JWT sessions — no passwords stored in application code
- Row Level Security on all 20 tables — data access enforced at the database level, not only the frontend
- Gmail App Password — scoped, revocable 16-character credential; real account password never exposed
- Groq API key stored as a Vite environment variable — not committed to version control
- Gmail credentials stored as Supabase Edge Function secrets — never exposed to the browser
- SECURITY DEFINER helper function (is_channel_member) — prevents RLS infinite recursion on chat membership queries without opening up the table to unscoped access
- Feature-branch git workflow — production (main branch / Netlify) never touched during active development
- All file uploads path-sanitized before storage to prevent path traversal

15. Deployment Guide

15.1 GitHub Codespaces (Wipro Laptop — No Installation)

- Go to: <https://github.com/Architnaman/newsroom-portal>
- Click Code > Codespaces > Create codespace on feature-development
- In terminal: `cd newsroom-ui`
- Create `.env` file with cloud Supabase credentials (see Section 15.3)
- Run: `npm install`
- Run: `npm run dev -- --host 0.0.0.0`
- Open Ports tab > port 5173 > click globe icon to open in browser (HTTPS URL — required for camera/video/audio)

15.2 Personal Laptop (Local Supabase)

- Start Supabase: `cd reporter_rostering_project && supabase start`
- Start edge functions: `supabase functions serve --env-file supabase/functions/.env`
- In new terminal: `cd newsroom-ui && npm run dev`
- Open browser: `http://localhost:5173` (camera works on localhost without HTTPS)

15.3 Environment Variables

Variable	Value
VITE_SUPABASE_URL	<code>https://vhremychmjzpunymvopt.supabase.co</code>
VITE_SUPABASE_ANON_KEY	<code>sb_publishable_zjwUc2uebOMUPEGsBorMpA_g0NK6f3O</code> (full key in <code>.env</code> file)
VITE_GROQ_API_KEY	<code>gsk_dXQsIRnTr2HMZSRNgmnDWGdyb3FY...</code> (full key in <code>.env</code> file)

16. Theme and Appearance System

Feature	Options	Storage
Color Mode	Light / Dark	localStorage (nr_theme)
Font Size	A- / A / A+ / A++ (4 sizes)	localStorage (nr_fontsize)
Background	8 presets: Default, Deep Blue, Purple, Forest, Midnight, Ocean, Sunset, Emerald	localStorage (nr_background)
Date Format	7 formats set by Admin	Supabase app_settings table
Week Start	Monday / Sunday set by Admin	Supabase app_settings table

All settings are persisted in localStorage and applied instantly across the app on load. Admin date format and week start day propagate to all open portals in real time via Supabase subscriptions with no page refresh required.

17. Project Outcome

The Newsroom Portal delivers a production-ready, role-based editorial management system that replaces manual coordination (WhatsApp messages, spreadsheets, email chains) with a structured, data-driven workflow. Key outcomes:

- Editors assign stories to the best-matched available reporter in seconds using the AI scoring engine, replacing manual availability and beat checking
- Override assignment disputes are tracked with full audit trails rather than being informal verbal agreements
- Leave requests have a formal approval workflow with automatic availability updates and email confirmation
- Ambient Scribe eliminates manual post-meeting assignment emails by extracting assignment decisions directly from meeting recordings or transcripts
- The 9-type notification system ensures every assignment, leave decision, and story status change reaches the right person automatically across all workflow surfaces
- Reporters can submit video field reports with professional trim and merge controls directly from their mobile device without any external app
- The analytics dashboards give management visibility into editorial output (story throughput, deadline performance, reporter workload) and app adoption (page popularity, user drop-off, per-user activity)
- The real-time chat system replaces informal messaging with a structured, searchable, archived communication layer inside the same portal